

VIVEK KUMAR RAI

Software Developer | Full-Stack Developer | Backend Developer

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PROFESSIONAL SUMMARY

Software Developer / Software Engineer with hands-on experience in full-stack and backend development using JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js, REST APIs, and SQL/NoSQL databases (MongoDB, PostgreSQL, MySQL). Experienced in Agile development, version control (Git), unit testing, CI/CD-style deployment workflows, cloud deployment (Vercel, Render, Supabase), authentication, API integrations, database workflows, and AI-powered features using LLM integration (OpenAI, Gemini, Groq). Strong foundation in Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, and the Software Development Life Cycle (SDLC).

TECHNICAL SKILLS

Languages: C++, JavaScript (ES6+), TypeScript, SQL

Frontend: HTML5, CSS3, React.js, Next.js 14, Tailwind CSS

Backend: Node.js, Express.js, RESTful APIs, JWT, API Rate Limiting

Databases: MongoDB, PostgreSQL, MySQL, Query Optimization

AI and APIs: OpenAI API, Gemini, Groq API, LLM Integration

Tools and Cloud: Git, GitHub, Version Control, Postman, Supabase, Vercel, Render, Cloud Deployment

Practices: Agile Development, Scrum, Unit Testing, Debugging, CI/CD-style Deployment Workflows, Code Review

Core CS: Data Structures and Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, SDLC, System Design (Basic)

WORK EXPERIENCE

Full-Stack Development Intern, Coding Arena

May 2026 – Present | Remote

- Contributed to CampusWire, a campus publishing and community platform built with Next.js 14, React, TypeScript, Express.js, Supabase, and PostgreSQL.
- Developed backend REST APIs for posts, comments, voting, fact-checking, reporting, profiles, media uploads, and moderation workflows.
- Implemented authentication and authorization using institutional email OTP, Row Level Security (RLS), JWT claims, and protected routes.
- Built secure content workflows with database-backed rate limiting, duplicate detection, URL blacklist checks, and audit logging.
- Integrated Groq-based text moderation and Sightengine image moderation, routing flagged content to an admin moderation queue.

PROJECTS

CodeGuardianAI — AI-Powered Code Security Platform

React.js, Node.js, OpenAI, Gemini, Groq API, Monaco Editor | March 2026

- Developed an automated AI code-review engine achieving 95% precision in vulnerability detection across 12+ programming languages.
- Integrated multiple LLM APIs into a unified code-analysis workflow, reducing manual security-auditing effort by 40%.
- Identified 50+ critical security vulnerabilities during testing and improved the reliability of the security-analysis workflow.

Tiffin Delivery Management System — Full-Stack Platform

React.js, Node.js, MongoDB, JWT, Razorpay | March 2026

- Engineered an end-to-end delivery platform supporting 500+ transactions through Razorpay payment integration.
- Implemented JWT-based authentication and granular access control for 100+ users.
- Streamlined order-processing workflows and user notifications, improving operational speed by 30%.

EDUCATION

Sagar Institute of Research and Technology (SIRT)

B.Tech in Computer Science & IT | 2023 – 2027 | CGPA: 8.04

Relevant Coursework: Data Structures and Algorithms, Operating Systems, DBMS, Computer Networks, System Design (Basic)

ACHIEVEMENTS & CERTIFICATIONS

- Solved 250+ Data Structures and Algorithms problems on LeetCode, spanning arrays, trees, graphs, and dynamic programming.
- International participant at SUNHACKS 2K25.
- Certification: Deloitte Data Analytics Job Simulation.
- Certification: Data Science with Python (NPTEL).